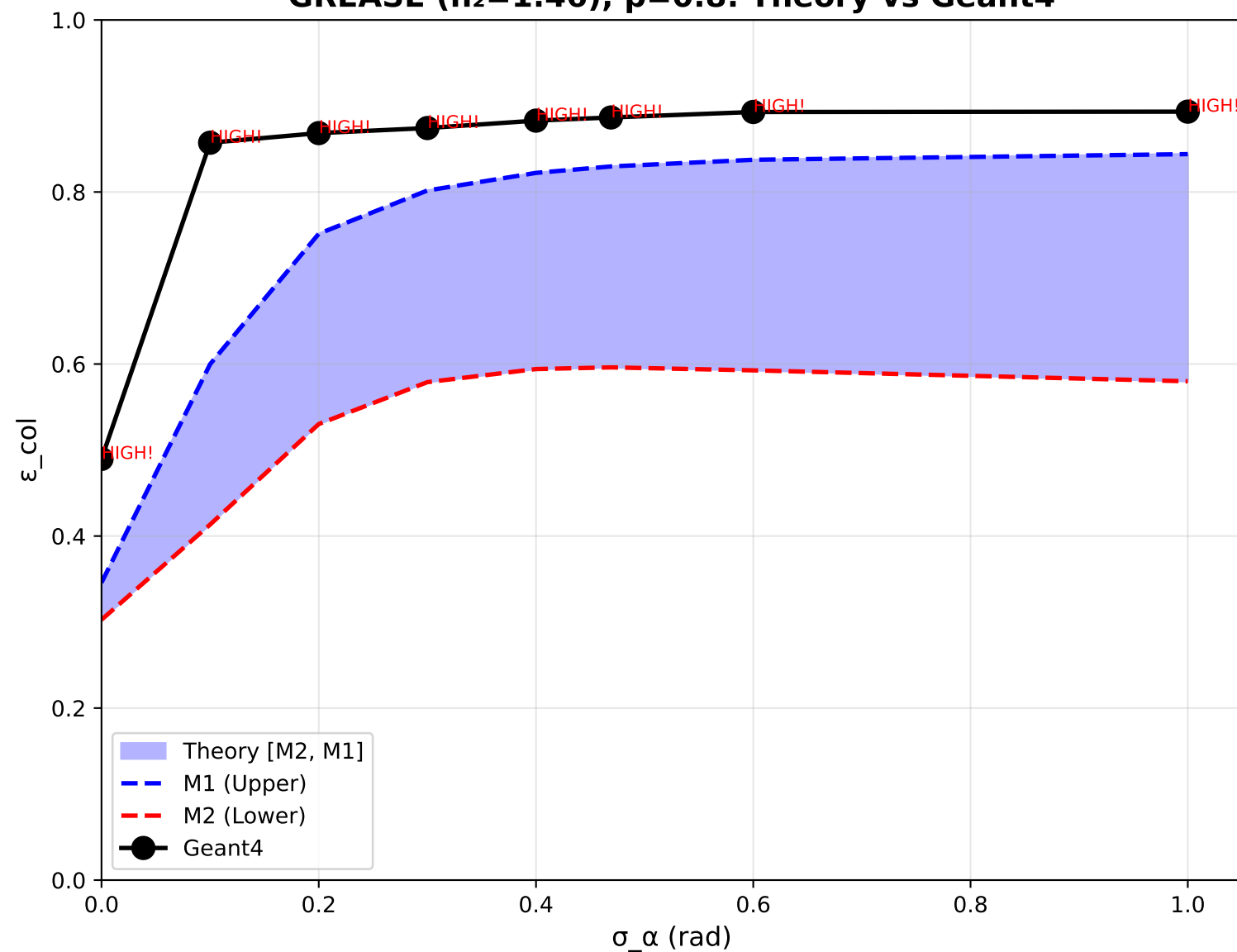
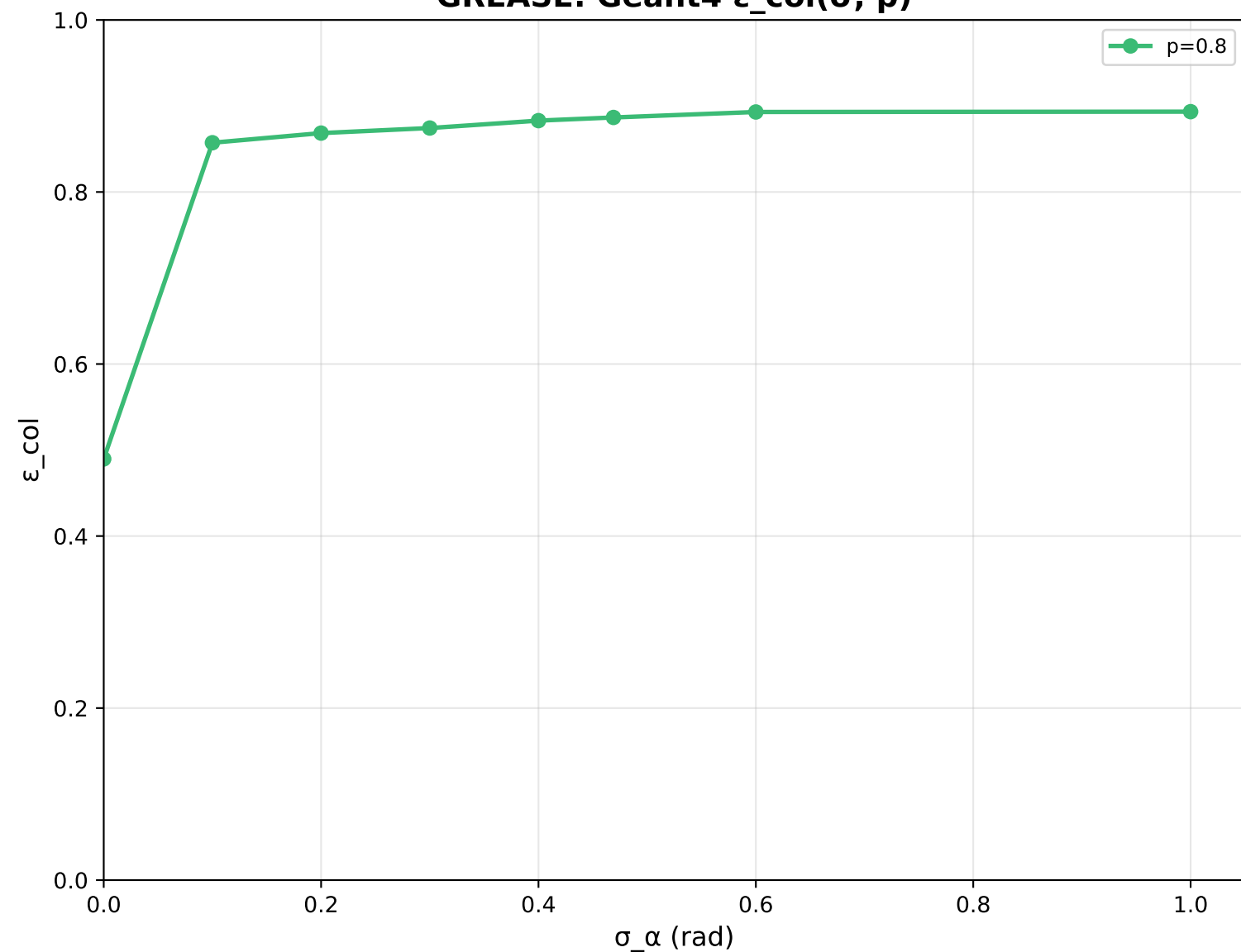


Theory Validation: Geant4 vs Closed-form Model ($n_1=1.86$, $R_c=0.975$, $p_b=1/6$)

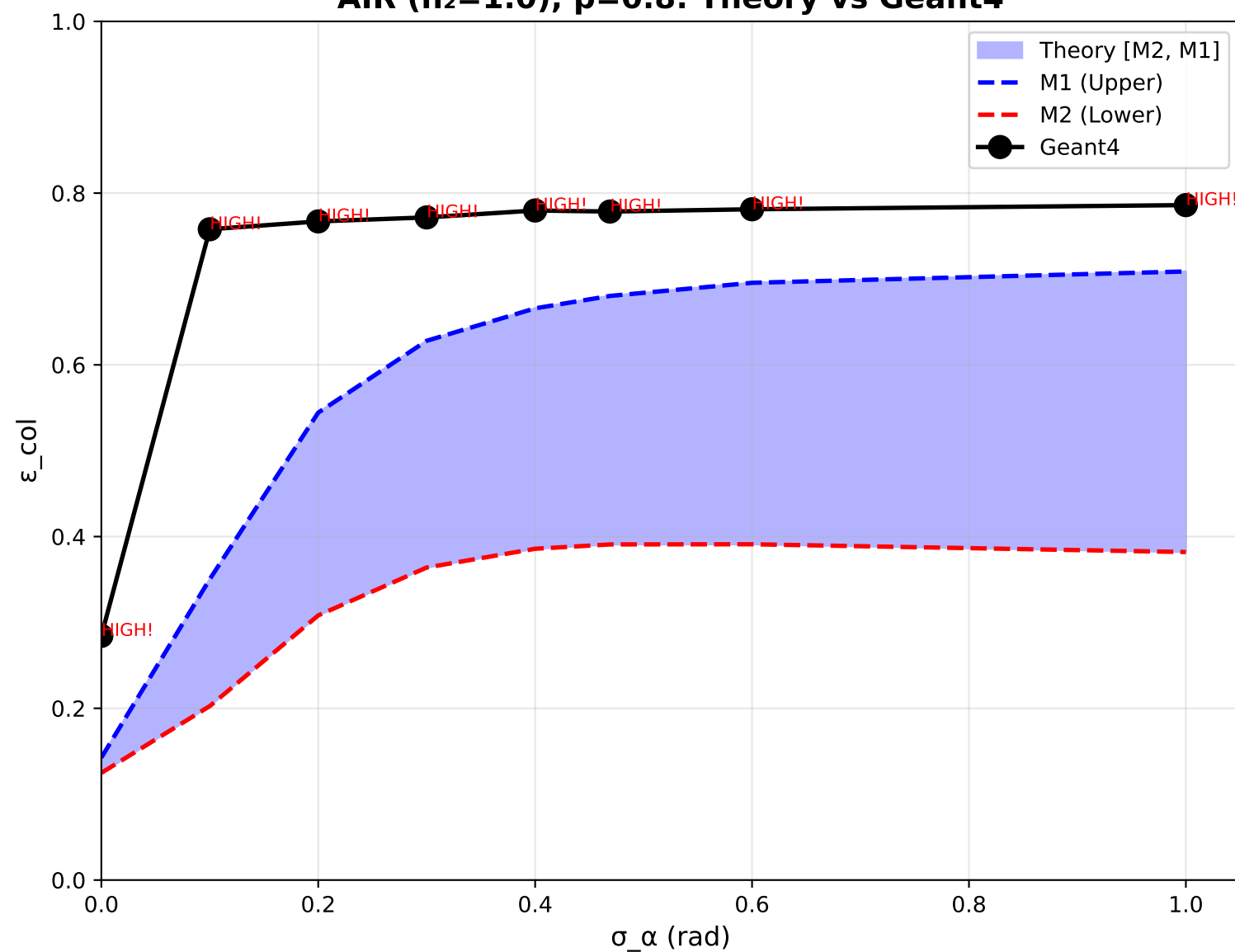
GREASE ($n_2=1.46$), $p=0.8$: Theory vs Geant4



GREASE: Geant4 $\varepsilon_{col}(\sigma, p)$



AIR ($n_2=1.0$), $p=0.8$: Theory vs Geant4



AIR: Geant4 $\varepsilon_{col}(\sigma, p)$

